

Mobile Development

(Lab 6)

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March 1st, 2022



Smart Software System Team



“Android is one of the most open systems I've ever seen. What makes Android great is it's literally designed from the ground up to be customized in a very powerful way..”

– Sundar Pichai – Google director



Learning Objectives

- Provide you with realistic, hands-on experience developing Android applications.
- Create a portfolio of apps that you can show your friends, discuss in interviews, and borrow for other applications.

Install guide



- Access these links and install OpenCV SDK:

https://docs.opencv.org/2.4/doc/tutorials/introduction/android_binary_package/O4A_SDK.html

- https://docs.opencv.org/2.4/doc/tutorials/introduction/android_binary_package/dev_with_OCV_on_Android.html

Exercise



- Access these links, implement face detection app using your face:

<https://opencv.org/face-recognition-on-android-using-openvino-toolkit-with-arm-plugin/>

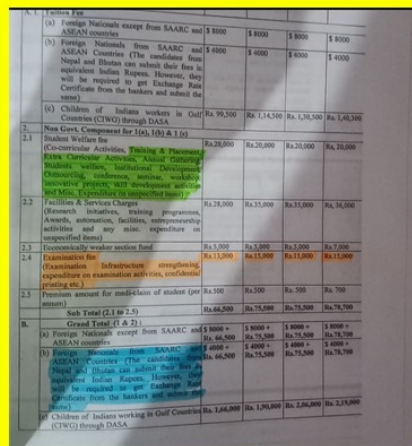
- Download this file and follow these instructions:

<https://drive.google.com/file/d/11LdWH9iDLEldRPLHBLg7KeP62Kdb23tW/view?usp=sharing>

Exercise

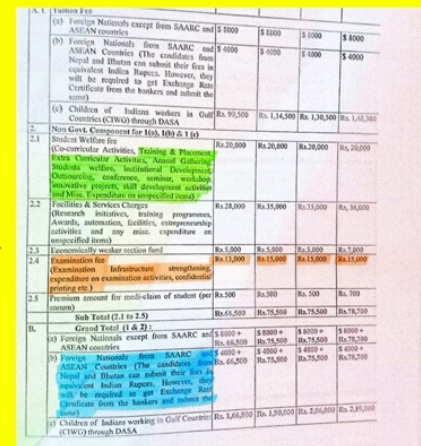
- Do app like this
- Take your favorite book with shadow and do a function to remove it with open-cv
- Note: If your project have already used OpenCV, you can submit it instead.

AndroidDocumentFilter



A photograph of a document page with a prominent shadow cast on its right side. The document contains a table with multiple columns and rows of text and numbers. The shadow is dark and obscures some of the content on the right edge of the page.

Original Image



The same document page as the original image, but with the shadow removed. The text and numbers are now clearly visible across the entire width of the page. The background is a uniform light color.

Shadow Removal

Exercise – Hint – Shadow removal



```
8
9  import org.opencv.android.Utils;
10 import org.opencv.core.Core;
11 import org.opencv.core.CvType;
12 import org.opencv.core.Mat;
13 import org.opencv.imgproc.Imgproc;
14
15 import java.util.ArrayList;
16 import java.util.List;
17 import java.util.concurrent.Executor;
18 import java.util.concurrent.Executors;
19
20 public class ShadowRemovalFilter {
21
22     public MyCallback callBack;
23
24     public interface MyCallback<Bitmap>{
25         void onComplete(Bitmap bitmap);
26     }
27
28     public static void getShadowFilteredImage(Bitmap bit_map, final ShadowRemovalFilter
29
30         Clahe.getClaheImage(bit_map, new Clahe.Callback<Bitmap>() {
31             @Override
32             public void onComplete(final Bitmap pre_processed) {
```

```
Executor executor = Executors.newSingleThreadExecutor();
final Handler handler = new Handler(Looper.getMainLooper());
executor.execute(new Runnable() {
    @Override
    public void run() {
        Mat srcArray = new Mat(pre_processed.getWidth(), pre_processed.getHeight(), CvType.CV_8UC1);
        Utils.bitmapToMat(pre_processed, srcArray);

        Imgproc.cvtColor(srcArray, srcArray, Imgproc.COLOR_BGR2HSV);

        List<Mat> bgrPlanes = new ArrayList<>();
        List<Mat> list = new ArrayList<>();
        List<Mat> result = new ArrayList<>();

        Core.split(srcArray, bgrPlanes);
        list.add(bgrPlanes.get(2)); // adding the V channel for processing in list

        result.add(new Mat(0, bgrPlanes.get(0))); // adding H channel in result
        result.add(new Mat(1, bgrPlanes.get(1))); // adding S channel in result

        // processing the V channel for shadow removal
        for (Mat mat : list) {
            Mat dilated_img = new Mat();
            Mat kernel = Mat.ones(7, 7, CvType.CV_32F);
            Imgproc.dilate(mat, dilated_img, kernel);
            Imgproc.medianBlur(dilated_img, dilated_img, 21);
            Mat diff = new Mat();
            Core.absdiff(mat, dilated_img, diff);
            Core.bitwise_not(diff, diff);
            Mat norm = diff.clone();
            Core.normalize(diff, norm, alpha: 0, beta: 255, Core.NORM_MINMAX, CvType.CV_8UC1);
            result.add(norm); // completely processed --> adding V channel into result
        }

        Mat result_norm = new Mat();
        Core.merge(result, result_norm);
```

Exercise – Hint – Shadow Removal



```
Mat result_norm = new Mat();
Core.merge(result, result_norm);

Imgproc.cvtColor(result_norm, result_norm, Imgproc.COLOR_HSV2BGR); // converting image back to RGB

final Bitmap res_ult = Bitmap.createBitmap(srcArray.cols(), srcArray.rows(), Bitmap.Config.ARGB_8888);
Utils.matToBitmap(result_norm, res_ult);

handler.post(new Runnable() {
    @Override
    public void run() {
        callBack.onComplete(res_ult);
    }
});
}
```


Exercise – Hint – Main



```
package com.namangarg.androiddocumentscanandfilter;

import ...

public class MainActivity extends AppCompatActivity {

    private static final int IMAGE_REQUEST = 1;
    DocumentFilter documentFilter;
    Button load, g1, g2, g3, g4, g5;
    ImageView imageView;
    Bitmap gb1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        load = findViewById(R.id.load);
        g1 = findViewById(R.id.g1);
        g2 = findViewById(R.id.g2);
        g3 = findViewById(R.id.g3);
        g4 = findViewById(R.id.g4);
        g5 = findViewById(R.id.g5);
        imageView = findViewById(R.id.image);

        documentFilter = new DocumentFilter();
    }
}
```

```
g5.setOnClickListener((v) -> {
    documentFilter.getShadowRemoval(gb1, (bitmap) -> {
        imageView.setImageBitmap(bitmap);
    });
});

load.setOnClickListener((v) -> {
    Intent intent = new Intent();
    intent.setType("image/*");
    intent.setAction(Intent.ACTION_GET_CONTENT);
    startActivityForResult(intent, IMAGE_REQUEST);
});

@Override
protected void onActivityResult(int requestCode, int resultCode, @Nullable Intent data) {
    super.onActivityResult(requestCode, resultCode, data);

    if (requestCode == IMAGE_REQUEST && resultCode == RESULT_OK) {
        Uri uri = data.getData();
        try {
            gb1 = MediaStore.Images.Media.getBitmap(this.getContentResolver(), uri);
        } catch (IOException e) {
            e.printStackTrace();
        }
        imageView.setImageURI(uri);
    }
}
```



References

- <https://web.stanford.edu/>
- <https://developer.android.com/>
- <https://introcs.cs.princeton.edu/>
- <https://www.cs.utexas.edu/>
- <https://www.codepath.com/>
- <https://geeksforgeeks.com/>
- proandroidev



Thank you for listening

*"Coming together is a beginning;
Keeping together is progress;
Working together is success."*
- HENRY FORD